

# Sequential Motion Planning for Bipedal Somersault via Flywheel SLIP and Momentum Transmission with Task Space Control

Xiaobin Xiong and Aaron D. Ames

**Abstract**—In this paper, we present a sequential motion planning and control method for generating somersaults on bipedal robots. The somersault (backflip or frontflip) is considered as a coupling between an axile hopping motion and a rotational motion about the center of mass of the robot; these are encoded by a hopping Spring-loaded Inverted Pendulum (SLIP) model and the rotation of a Flywheel, respectively. We thus present the Flywheel SLIP model for generating the desired motion on the ground phase. In the flight phase, we present a momentum transmission method to adjust the orientation of the lower body based on the conservation of the centroidal momentum. The generated motion plans are realized on the full-dimensional robot via momentum-included task space control. Finally, the proposed method is implemented on a modified version of the bipedal robot Cassie in simulation wherein multiple somersault motions are generated.

## I. INTRODUCTION

Human athletes perform somersaults as an ultimate demonstration of mobility and dynamic maneuverability. The anthropomorphic partner in the robotics world, the hydraulic-actuated robot Atlas from Boston dynamics, was able to backflip [1], showing the decades of dedicated effort towards advancing robotic technologies; this dates back to the frontflips of the Raibert’s hopper [2], [3] in the 1980s. Regardless of the practical ramifications of doing somersaults on bipedal robots, achieving this kind of highly dynamic behavior benchmarks the stage of maturity of engineering robotic systems.

The back-flip of Atlas inspired the quadrupedal locomotion community which recently has built and controlled electric-powered quadrupedal robots to perform backflips [4], [5]. However, these results have not been widely seen on full-scale bipedal robots, despite the existence of studies of bipedal somersaulting in [6], [7], [8].

This is in part due to the fact that bipedal robots typically have more complex designs and, roughly speaking, less region of stability comparing to quadrupeds. These somersault behaviors require robot design to be able to make the behaviors feasible so that feedback control can play a role. The first observation is that the actuators need to be powerful to generate large vertical acceleration so that the robot can jump up to a certain height. The angular acceleration should be large enough so that enough momentum is generated on the ground to enable whole-body rotation is the flight phase.

Another observation is that backflipping requires the robot to have an upper body with large inertia and large ranges of

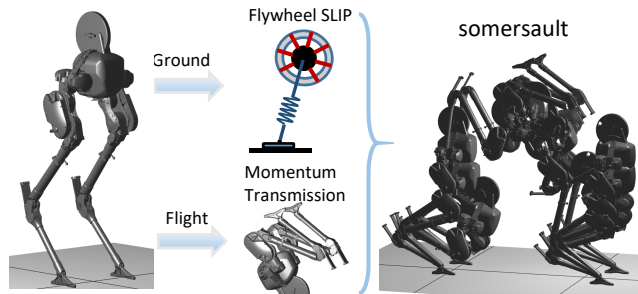


Fig. 1. Illustration of the proposed methodology for generating somersaults.

motion at certain joints. To achieve the flip motion, the robot should have enough momentum at lift-off, which should be mainly generated by the upper body. This is because the lower body typically extends for jumping, the momentum of which cannot be freely generated. Additionally, the robot should have large ranges of motion at joints that connect to the linkages with large inertia; for instance, if the hip joints have small ranges of motion, then it requires impractical torques to accelerate the joints to the required rotational velocity within the ranges of motion.

As illustrated thus far, achieving somersaults is a complex system-level problem of design and control. In this paper, we mainly investigate the motion planning and feedback control component in achieving these motions. The robot design to make somersaults feasible is out of the scope of this paper, even though it is an important and challenging problem.

In [9], [4], the motion plans for backflips are generated via trajectory optimization on the full-dimensional model of the quadrupeds, which has computational challenges for more complex bipedal robots. The formulated optimization is also highly nonconvex in a large dimensional space [10], [11], [12]. The optimization typically requires good initial guesses to solve, which can have feasibility issues. If the robot design renders the motion infeasible to realize, the trajectory optimization runs into infeasibility and provides little information towards the possible improvement in the robot design. Moreover, the optimized plans based on the full model typically are not robust to modeling errors and external disturbances. Thus a principled simple representation, e.g. the Linear Inverted Pendulum (LIP) [13] for walking and the Spring-loaded Inverted Pendulum (SLIP) [14] for running, is preferred for studying the essential components which realize the somersault.

To address this, we present a Flywheel SLIP (FSLIP)

# 基于飞轮SLIP与动量传递及任务空间控制的序列运动规划实现双足空翻

Xiaobin Xiong 与 Aaron D. Ames

**摘要**——本文提出了一种序列运动规划与控制方法，用于在双足机器人上生成空翻动作。空翻（后空翻或前空翻）被视为机器人轴向跳跃运动与绕质心旋转运动的耦合；这两种运动分别由跳跃式弹簧负载倒立摆（SLIP）模型和飞轮旋转来表征。因此，我们提出了飞轮SLIP模型，用于在地面阶段生成期望运动。在飞行阶段，我们提出了一种基于质心动量守恒的动量传递方法，以调整下半身的姿态。所生成的运动规划通过含动量的任务空间控制在全维机器人上实现。最后，该方法在改进版双足机器人Cassie的仿真中得以实施，并生成了多种空翻动作。

## I. 引言

人类运动员通过完成空翻，展现了极致的机动性与动态操控能力。在机器人领域，波士顿动力公司（Boston Dynamics）的液压驱动机器人Atlas成功实现了后空翻 [1]，这标志着数十年来推动机器人技术发展的不懈努力——其渊源可追溯至20世纪80年代Raibert跳跃机器人 [2], [3] 的前空翻。尽管在双足机器人上实现空翻的实际意义尚待探讨，但达成此类高度动态的行为，无疑是对工程机器人系统成熟度的标杆性检验。

Atlas的后空翻激励了四足运动研究领域，该领域近期已构建并控制电动四足机器人完成了后空翻 [4][5]。然而，尽管在 [6], [7], [8]中已有关于双足空翻的研究，这些成果尚未在全尺寸双足机器人上广泛实现。

这部分是由于双足机器人通常设计更为复杂，且粗略而言，其稳定区域小于四足机器人。这些空翻行为要求机器人设计能够使行为可行，从而使反馈控制得以发挥作用。首要观察是，执行器需要具备强大动力以产生较大的垂直加速度，从而使机器人能够

跳跃至一定高度。角加速度应足够大，以便在地面上产生足够的动量，从而在飞行阶段实现全身旋转。

另一个观察结果是，后空翻要求机器人拥有一个具有大惯性且某些关节运动范围较大的上半身。

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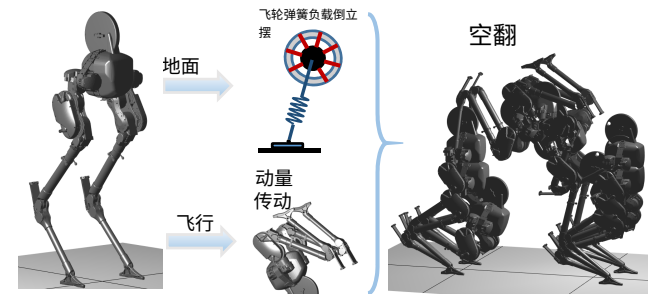


图1. 所提出的空翻生成方法示意图。

为了实现翻转运动，机器人在离地时应具备足够的动量，这主要应由上半身产生。这是因为下半身通常为跳跃而伸展，其动量无法自由生成。此外，机器人连接大惯性连杆的关节应具有较大的运动范围；例如，如果髋关节的运动范围较小，则需要不切实际的力矩才能在运动范围内将关节加速到所需的旋转速度。

如上所述，实现空翻是一个涉及设计与控制的复杂系统级问题。本文主要研究实现这些动作的运动规划与反馈控制部分。使空翻可行的机器人设计不在本文讨论范围内，尽管这是一个重要且具有挑战性的问题。

在 [9], [4]中，后空翻的运动规划是通过将四足机器人的全维模型进行轨迹优化生成的，这对于更复杂的双足机器人而言存在计算挑战。所构建的优化问题在大维度空间 [10][11], [12]中也高度非凸。该优化通常需要良好的初始猜测才能求解，这可能导致可行性问题。如果机器人设计使得动作无法实现，轨迹优化将陷入不可行状态，并且无法为机器人设计的可能改进提供有效信息。此外，基于全模型的优化规划通常对建模误差和外部干扰缺乏鲁棒性。因此，一种有原则的简化表示方法，例如用于行走的线性倒立摆 [13]和用于奔跑的弹簧负载倒立摆 [14]，更适合用于研究实现空翻的关键要素。

为解决这一问题，我们提出了一种飞轮SLIP (FSLIP)

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model to capture somersault dynamics on the ground phase, including jumping and landing phases. The FSLIP has a flywheel [15], [16], [17] which is attached on a SLIP [18] model. The point mass dynamics of the FSLIP represents the hopping part of the somersault, while the flywheel captures the centroidal angular dynamics [19]. Trajectory optimization is formulated on the FSLIP to generate jumping and landing behaviors subject to the task constraints and additional physical constraints of the somersault. With the low dimensionality, it is easy and fast to solve the optimization.

When the robot is in the flight phase, the centroidal angular momentum [20], [21], [22], [23], [24] is conserved. We present a moment transmission control for a specific linkage to modulate the angle of the rest of the body. The specific linkage can be a flywheel, a tail, or arms of the robot. Moreover, additional assistive strategies are presented towards the landing.

The motion planning from the reduced-order representation is generated in sequence to the domains of the somersault. The motion plan is realized on the full-dimensional robot via an optimization-based controller. The optimization-based controller also takes into consideration the ground reaction force constraints and the input constraints.

The proposed method is realized on a modified version of the bipedal robot Cassie in simulation. We added a flywheel on the top of its pelvis to increase the inertia of its upper body and to amplify its capability to control the momentum. The flywheel can also be viewed as a representation of arms on the robot that have a free range of motion at the shoulders. Various somersault motions are generated, including front/back-flips and frontflips with over 1m forward jumping distance. Robustness and practical feasibility are also analyzed and discussed. With the formulated FSLIP model and the correspondence to the realization on a full dimensional robot, we hope that this study can shed some light on bipedal robot design and motion generation towards realizing versatile aerial motion on bipedal robots.

## II. MODELLING

We start by presenting the model of the robot with the modification. Then we model the hybrid dynamics of the somersault, which is used in the simulation and analysis of the proposed planning and control methods.

### A. Robot Model

The robot Cassie is a bipedal robot that and its predecessor ATRIAS [25] are built to best mimic the canonical Spring-loaded Inverted Pendulum (SLIP) model. As a result, it has a concentrated upper body and mechanical springs in the leg. The robot has shown good mechanical properties for walking [26], [27]. However, the trivial torso on Cassie makes it extremely difficult to produce any movement with aerial phases.

For instance, in [28], vertical hopping is realized on the robot with a goal of producing zero momentum before the flight phase. The leg extension generates a non-zero angular momentum, and thus the torso has to cancel out the angular

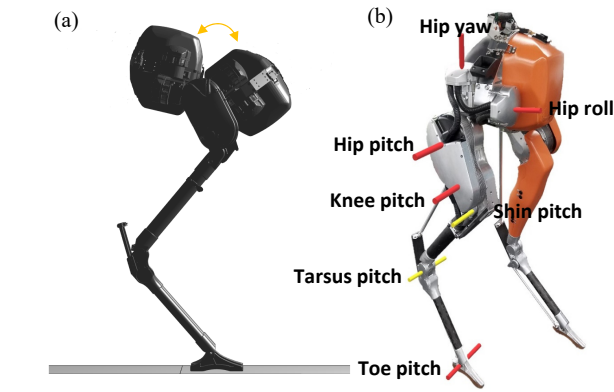


Fig. 2. (a) The range of motion of the pelvis. (b) The joints on the lower body of the bipedal robot Cassie by Agility Robotics.

momentum by producing a large forward rotation of the pelvis, which is limited by its range of motion (see Fig. 2 (a)), and maximum speed at the hip pitch joint. In other words, with a small torso to generate a large momentum, the required torso rotational speed is impractical to be realized within the range of motion. Thus using the trivial torso to generate somersaults also becomes impractical. To enable the capability, we present the following modification.

**Hardware Modification:** A flywheel is added on the top of the pelvis of the robot Cassie. It can be viewed as a robotic tail in some sense, e.g. the tail on Salto [29]. The design can be optimized for the location and size of the flywheel. Since the focus of this paper is on the planning and control, we manually select the installation of the flywheel based on intuition. The flywheel is installed 15cm above the pelvis, assuming the flywheel is made of aluminum. The exact specifications of the flywheel are discussed in section VI. Except the flywheel, all physical parameters of the robot remain unchanged.

**Dynamics Model:** We use a model of 22 degrees of freedom (DOFs) to represent the robot Cassie [28]. With the flywheel modification, the robot has 23 DOFs. Let  $q \in SE(3) \times \mathbb{R}^{17}$  represent the configuration of the robot. The Euler-Lagrange equation for its dynamics is:

$$M(q)\ddot{q} + H(q, \dot{q}) = Bu + J_s^T \tau_s + J_{h,v}^T F_{h,v}, \quad (1)$$

$$J_{h,v}(q)\ddot{q} + \dot{J}_{h,v}(q)\dot{q} = 0, \quad (2)$$

where  $M(q)$  is the mass matrix,  $H(q, \dot{q})$  is the Coriolis, centrifugal and gravitational term,  $B$  and  $u \in \mathbb{R}^{11}$  are the actuation matrix and the motor torque vector,  $\tau_s$  and  $J_s$  are the spring joint moment vector and the corresponding Jacobian, and  $F_{h,v} \in \mathbb{R}^{n_{h,v}}$  and  $J_{h,v}$  are the holonomic force vector and the corresponding Jacobian respectively. The subscript  $v$  is used to indicate different domains which have different numbers of holonomic constraints.

### B. Hybrid Model of Somersaulting

The somersault is a hybrid dynamical phenomenon with ground and flight phases. We assume that the robot hops in the sagittal plane and its left and right feet have the same

model, 用于捕捉地面阶段（包括跳跃和着陆阶段）的空翻动力学。FSLIP 模型包含一个飞轮 [15], [16], [17], 该飞轮附着在 SLIP [18]模型上。FSLIP 的质点动力学代表了空翻的弹跳部分, 而飞轮则捕捉了质心角动力学 [19]。在 FSLIP 上构建了轨迹优化, 以生成满足任务约束和空翻额外物理约束的跳跃与着陆行为。由于维度较低, 该优化求解起来既简单又快速。

当机器人处于飞行阶段时, 质心角动量 [20], [21], [22], [23], [24] 守恒。我们提出了一种针对特定连杆的力矩传递控制方法, 以调节身体其余部分的角度。该特定连杆可以是机器人的飞轮、尾巴或手臂。此外, 还提出了额外的辅助策略以辅助着陆。

来自降阶表示的运动规划按顺序生成, 以对应空翻的各个域。该运动规划通过基于优化的控制器在全维机器人上实现。该基于优化的控制器还考虑了地面反作用力约束和输入约束。

所提出的方法在仿真中于双足机器人Cassie的改进版本上实现。我们在其骨盆顶部增加了一个飞轮, 以增加其上身的惯性, 并增强其控制动量的能力。该飞轮也可视为机器人手臂的一种表示, 这些手臂在肩部具有自由运动范围。生成了多种空翻运动, 包括前空翻/后空翻以及跳跃距离超过1米的前空翻。还分析和讨论了鲁棒性和实际可行性。通过构建的FSLIP模型及其在全维机器人上实现的对应关系, 我们希望这项研究能为双足机器人的设计和运动生成提供一些启示, 以实现在双足机器人上完成多样化的空中运动。

## II. 建模

我们首先介绍经过修改的机器人模型。然后, 我们对空翻的混合动力学进行建模, 该模型将用于所提出的规划与控制方法的仿真与分析。

### A. 机器人模型

机器人 Cassie 是一款双足机器人, 其前身 ATRIAS [25] 的设计旨在最大限度地模仿经典的弹簧-负载倒立摆 (SLIP) 模型。因此, 它具有集中的上半身和腿部的机械弹簧。该机器人在行走方面表现出良好的机械特性[26], [27]。然而, Cassie上简单的躯干结构使得产生任何带有腾空阶段的运动变得极其困难。

例如, 在 [28], 中, 机器人在飞行阶段前以实现零动量为目标完成垂直跳跃。腿部伸展会产生非零角动量, 因此躯干必须通过骨盆的大幅前向旋转来抵消该角动量, 而这一旋转受到其运动范围 (见图2 (a) ) 以及髋关节俯仰关节最大速度的限制。

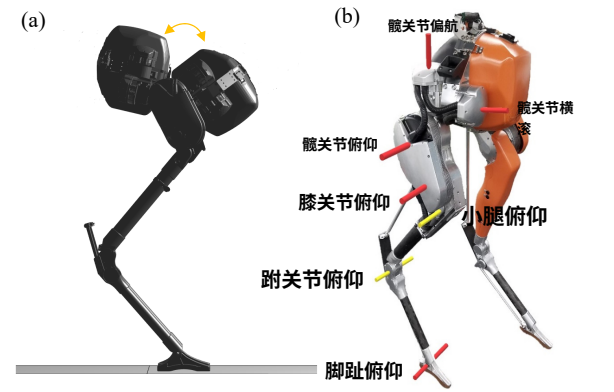


图2. (a) 骨盆的运动范围。 (b) Agility Robotics 的双足机器人 Cassie 下半身的关节。

换言之, 若利用小型躯干产生大动量, 所需的躯干转速在实际运动范围内难以实现。因此, 利用普通躯干完成空翻也变得不切实际。为赋予机器人这一能力, 我们提出以下改进方案。

**硬件改造:** 在机器人 Cassie 的骨盆顶部增加了一个飞轮。从某种意义上, 它可以被视为一个机器人尾巴, 例如 Salto 上的尾巴 [29]。可以针对飞轮的位置和尺寸对设计进行优化。由于本文的重点是规划与控制, 我们根据直觉手动选择了飞轮的安装位置。飞轮安装在骨盆上方15厘米处, 假设飞轮由铝制成。飞轮的具体规格在第VI节中讨论。除飞轮外, 机器人的所有物理参数保持不变。

动力学模型: 我们使用一个具有22个自由度的模型来表示机器人 Cassie [28]。经过飞轮改造后, 机器人具有23个自由度。令  $q \in SE(3) \times \mathbb{R}^{17}$  表示机器人的构型。其动力学的欧拉-拉格朗日方程如下:

$$M(q)\ddot{q} + H(q, \dot{q}) = Bu + J_s^T \tau_s + J_{h,v}^T F_{h,v}, \quad (1)$$

$$J_{h,v}(q)\ddot{q} + \dot{J}_{h,v}(q)\dot{q} = 0, \quad (2)$$

其中  $M(q)$  是质量矩阵,  $H(q, \dot{q})$  是科里奥利、离心和重力项,  $B$  而  $u \in \mathbb{R}^{11}$  是驱动矩阵和电机扭矩向量,  $\tau_s$  而  $J_s$  是弹簧关节力矩向量及其对应的雅可比矩阵,  $F_{h,v} \in \mathbb{R}^{n_{h,v}}$  而  $J_{h,v}$  分别是完整约束力向量及其对应的雅可比矩阵。下标  $v$  用于表示具有不同数量完整约束的不同域。

### B. 空翻混合模型

空翻是一种混合动力学现象, 包含地面阶段和飞行阶段。我们假设机器人在矢状面内跳跃, 且其左右足部具有相同的

contact mode. For simplification, we also assume the feet always flatly contact the ground, which will be encoded as one of the control objectives. Based on the sequence of the contact modes, we divide the motion into three domains, i.e.,  $v = \text{jumping, flight, and landing}$ . In jumping, the feet are in contact with the ground. It transits from jumping to flight when the ground reaction normal forces reach to zero. In the flight phase, the feet are above the ground. The robot goes through a large whole-body rotation in its sagittal plane. It transits into landing when the feet strike the ground. The foot-ground impact is modeled as plastic impact [30], where the position trajectories are smooth while the velocities are discontinuous.

### III. THE FLYWHEEL SLIP MODEL

In this section, we explain the Flywheel SLIP (FSLIP) model for generating the jumping and landing trajectories for the robot. The point mass dynamics of the FSLIP captures the hopping dynamics of the center of mass (COM) dynamics of the robot; the rotational dynamics of the flywheel corresponds to the rotational part of the centroidal dynamics of the robot. Using parametric regressions on the stiffness of the leg spring and the centroidal inertia of the robot, we identify the Flywheel SLIP model to best capture the features of the robot.

#### A. The Dynamics

The Flywheel SLIP is the actuated SLIP model in [18], [31] with a flywheel attached on the mass and an actuated foot (see Fig. 3 (a)). The actuated SLIP is the canonical SLIP model with leg length actuation and a prismatic spring. The flywheel is actuated to rotate about the mass location. The foot actuation is added to work with the flywheel to generate appropriate acceleration. The dynamics are given by:

$$m\ddot{x} = -\frac{\tau + u}{r}\cos(\beta) + F^s\sin(\beta) \quad (3)$$

$$m\ddot{z} = -mg + \frac{\tau + u}{r}\sin(\beta) + F^s\cos(\beta) \quad (4)$$

$$I\ddot{\theta} = \tau \quad (5)$$

$$\dot{s} = \dot{L} - \dot{r} \quad (6)$$

where:

- $m$  is the mass of the mass and the flywheel,
- $I$  is the rotational inertia of the flywheel,
- $\tau$  is the torque on the flywheel,
- $u$  is the foot actuation,
- $\theta$  is the angle of the flywheel,
- $\beta$  is the angle of the leg,
- $r$  is the real leg length,
- $L$  is the actuated leg length,
- $s$  is the spring deformation,
- $F^s$  is the spring force,
- $x, z$  are the horizontal and vertical mass positions.

The kinematic constraints are  $s + r = L$  and  $\dot{s} + \dot{r} = \dot{L}$ . The spring force is calculated by:  $F^s = Ks + D\dot{s}$ , where  $K, D$  are the stiffness and damping of the spring.

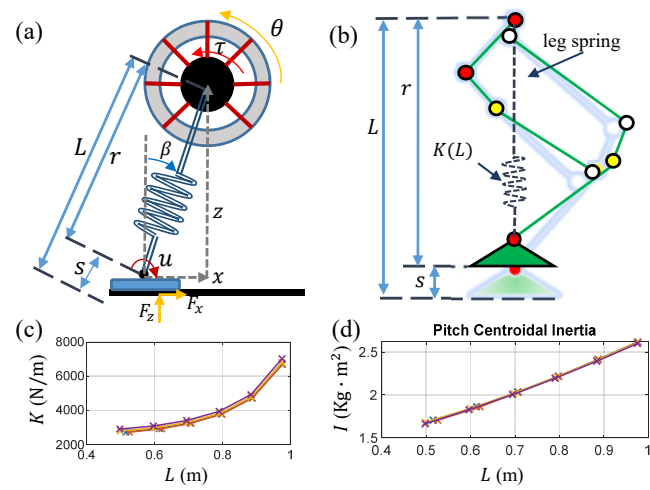


Fig. 3. (a) The Flywheel SLIP. (b) Illustration of the leg spring from the leg of Cassie. (c, d) The stiffness of the leg spring and the pitch centroidal inertia of the robot v.s. the leg length for different standing configurations.

**Correspondence to the robot:** The Flywheel SLIP model described above is general. Here we use parametric regressions to best match the model to the robot. Similar to [28], [18], we identify the stiffness and damping of the leg spring from the joint springs of the robot. Then parametric regressions are used to project the stiffness  $K$  and damping  $D$  of the leg spring over the leg length  $L$ . Thus the leg spring of the FSLIP mimics the leg of the robot, as illustrated by Fig. 3 (b). Additionally, we regress the centroidal pitch inertia [32]  $I(L)$  of the robot, which becomes the inertia of the Flywheel. It is important to note that it is the inertia of the whole robot instead of the flywheel on the pelvis that corresponds to the flywheel on the FSLIP. The regressed FSLIP model from the robot is expected to produce behaviors which can be translated to Cassie. Fig. 3 (c) (d) show the leg stiffness of the leg spring and the pitch inertia over different leg length for different configurations, indicating that they are mainly changed by  $L$ .

#### B. Jumping Optimization

Now we use the Flywheel SLIP model to generate a jumping motion from standing for enabling the somersault. The planned motion is required to satisfy the actuation constraints, the ground force constraints, the required flipping momentum, and the apex constraints. The planned motion contains the trajectories of the point mass and desired centroidal momentum, which will be tracked by the robot.

Similar to [28], we use a direct transcription method [33], [10] to describe and solve the optimization problem. The trajectories are discretized and the dynamics are integrated via trapezoidal integration. Here we mainly describe the constraints and the cost function of the optimization.

**Task Constraints:** The goal of the jumping optimization is to create enough vertical velocity and angular momentum to enable somersaults. The task is encoded via the velocities

接触模式。为简化起见，我们还假设足部始终与地面保持平坦接触，这将被编码为控制目标之一。基于接触模式的序列，我们将运动划分为三个域，即  $v = \text{跳跃、飞行和着陆}$ 。在跳跃阶段，足部与地面接触。当地面法向反力降至零时，从跳跃阶段过渡到飞行阶段。在飞行阶段，足部位于地面之上。机器人在其矢状面内经历大幅度的全身旋转。当足部撞击地面时，过渡到着陆阶段。足-地碰撞被建模为塑性碰撞 [30]，其中位置轨迹是平滑的，而速度是不连续的。

### III. 飞轮SLIP模型

本节将阐述用于生成机器人跳跃与着陆轨迹的飞轮SLIP (FSLIP) 模型。FSLIP的质点动力学捕捉了机器人质心 (COM) 动力学的弹跳特性；飞轮的旋转动力学则对应机器人质心动力学的旋转部分。通过对腿部弹簧刚度和机器人质心惯性进行参数回归，我们确定了最能表征机器人特征的飞轮SLIP模型。

#### A. 动力学

飞轮弹簧负载倒立摆 (Flywheel SLIP) 是在 [18], [31] 基础上增加了飞轮（安装于质量块上）和足部驱动的驱动SLIP模型（见图3(a)）。驱动SLIP模型是带有腿长驱动和棱柱弹簧的经典SLIP模型。飞轮被驱动以绕质量块位置旋转。足部驱动与飞轮协同工作，以产生适当的加速度。其动力学方程如下：

$$m\ddot{x} = -\frac{\tau + u}{r}\cos(\beta) + F^s\sin(\beta) \quad (3)$$

$$m\ddot{z} = -mg + \frac{\tau + u}{r}\sin(\beta) + F^s\cos(\beta) \quad (4)$$

$$I\ddot{\theta} = \tau \quad (5)$$

$$\dot{s} = \dot{L} - \dot{r} \quad (6)$$

其中：

- $m$  为质量块与飞轮的总质量，
- $I$  为飞轮的转动惯量，
- $\tau$  为作用于飞轮的扭矩，
- $u$  为足部驱动力，
- $\theta$  是飞轮角度，
- $\beta$  是腿角度，
- $r$  是实际腿长，
- $L$  是驱动腿长，
- $s$  是弹簧变形，
- $F^s$  是弹簧力，
- $x, z$  分别是水平和垂直质量位置。

运动学约束为  $s + r = L$  和  $\dot{s} + \dot{r} = \dot{L}$ 。弹簧力通过下式计算： $F^s = Ks + D\dot{s}$ ，其中  $K, D$  分别为弹簧的刚度和阻尼。

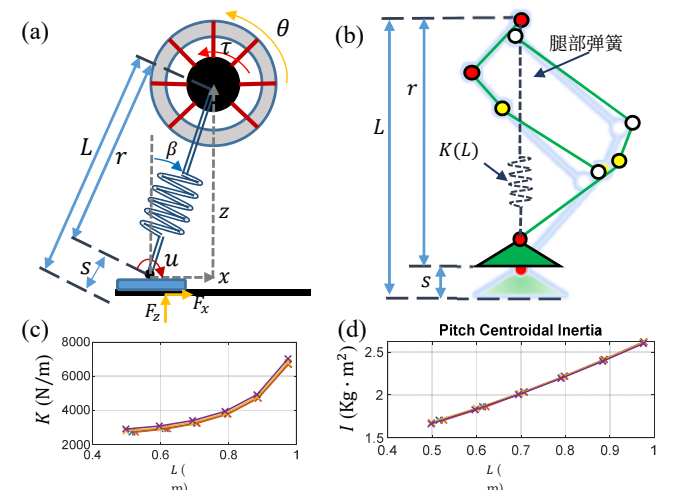


图3. (a) 飞轮SLIP。 (b) Cassie腿部弹簧示意图。 (c, d) 不同站立构型下腿部弹簧刚度及机器人俯仰质心惯性随腿长的变化。

**与机器人的对应关系：**上述飞轮SLIP模型具有通用性。此处我们采用参数回归方法，使模型与机器人实现最佳匹配。与[28], [18]类似，我们从机器人的关节弹簧中识别出腿部弹簧的刚度和阻尼。随后通过参数回归将腿部弹簧的刚度  $K$  和阻尼  $D$  投影到腿长  $L$  上。因此，FSLIP的腿部弹簧模拟了机器人的腿部，如图3(b)所示。此外，我们对机器人的质心俯仰惯量 [32]  $I(L)$  进行回归，该惯量成为飞轮的惯性。需要特别注意的是，与FSLIP飞轮对应的并非骨盆上的飞轮，而是整个机器人的惯性。从机器人回归得到的FSLIP模型预期能够产生可迁移至Cassie的行为。图3(c)(d)展示了不同构型下腿部弹簧刚度与俯仰惯量随腿长的变化，表明它们主要受  $L$  影响。

#### B. 跳跃优化

现在我们利用飞轮SLIP模型从站立状态生成跳跃运动，以实现空翻。规划的运动需满足驱动约束、地面力约束、所需翻转动量以及顶点约束。规划的运动包含质点轨迹和期望质心动量，这些将由机器人进行跟踪。

与 [28], 类似，我们采用直接转录方法 [33][10]来描述并求解优化问题。轨迹被离散化，动力学通过梯形积分进行积分。这里我们主要描述优化问题的约束条件和代价函数。

**任务约束：**跳跃优化的目标是产生足够的垂直速度和角动量以实现空翻。该任务通过离地事件时的速度进行编码。



at the lift-off event. We assume that the robot lands with a similar self-configuration to that at lift-off. Thus the duration of the flight phase is given by:

$$T_{\text{Flight}} = \frac{2\dot{z}^{\text{lift-off}}}{g}, \quad (7)$$

where  $\dot{z}^{\text{lift-off}}$  is the vertical velocity of the point mass at lift-off. If the robot fixes its configuration during the flight phase, the angular velocity of the whole body at lift-off should be:

$$\dot{\theta}^{\text{lift-off}} = \frac{\pm 2\pi - 2\beta^{\text{lift-off}}}{T_{\text{Flight}}} = \frac{(\pm\pi - \beta^{\text{lift-off}})g}{\dot{z}^{\text{lift-off}}}. \quad (8)$$

Note that we will relax the condition of the fixed configuration on the robot later. Eq. (8) establishes the constraint on the jumping phase to produce a flip motion. The sign  $\pm$  depends on if the somersault is a backflip or frontflip.

**Physical Constraints:** During jumping, we enforce non-slipping condition on the foot via,

$$-F_z\mu \leq F_x \leq F_z\mu, \quad (\text{Friction Cone})$$

where  $F_z$  is the vertical ground reaction force (GRF), and  $\mu$  is the friction coefficient. It is also desirable to keep the feet flat on the ground, thus the zero moment point (ZMP) constraint is included:

$$-F_z \frac{l}{2} \leq u \leq F_z \frac{l}{2}, \quad (\text{ZMP})$$

where  $l$  is the foot length. Additional constraints include the actuation limits, leg length limits and non-negative normal ground reaction forces. The final ground normal force should reach to 0 to trigger the lift-off event, i.e.,  $F_z^{\text{lift-off}} = 0$ .

**Cost:** The cost for the jumping optimization is on the virtually consumed energy:

$$J_{\text{Jumping}} = \sum_i (c_1 \ddot{L}_i^2 + c_2 u_i^2 + c_3 \tau_i^2) \Delta T, \quad (9)$$

where  $c_{1,2,3}$  are the coefficients,  $\Delta T$  is time discretization, and  $i$  is the index for the discretized variables. Fig. 4 illustrates an example of the optimized trajectories of a jumping behavior.

### C. Landing Optimization

The landing optimization is performed after the touch-down event at the end of the flight phase. Since it uses the same Flywheel SLIP model, we briefly explain the landing planning here.

During landing, the system rests to a static configuration from the flight phase. The task can be directly encoded via the equality constraints on the final states:

$$\begin{aligned} z_N &= z^{\text{des}}, \\ \dot{z}_N &= x_N = \dot{x}_N = \dot{\theta} = 0, \\ F_N^s &= mg, \end{aligned}$$

where the subscript  $N$  indicates the final state of landing. The physical constraints are the same to that of the jumping

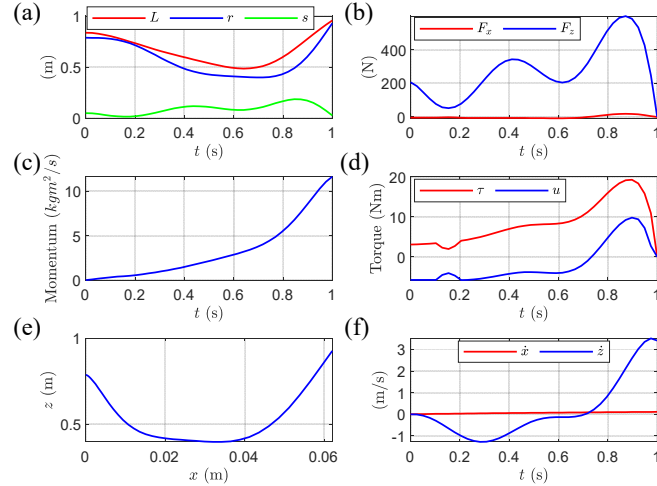


Fig. 4. An example of the optimized jumping trajectories of the leg length and spring deformation (a), ground reaction forces (b), the momentum of the flywheel (c), the torques (d), the position of the point mass (e) and the velocities of the mass (f).

planning. The initial states are constrained based on the post-impact state of the robot. The cost is the cost of jumping plus a penalty on the spring oscillation:

$$J_{\text{Landing}} = J_{\text{Jumping}} + \sum_i \alpha \dot{s}_i^2 \Delta T, \quad (10)$$

where  $\alpha$  is a weighting coefficient.

## IV. FLIGHT PHASE PLANNING

In this section, we present the planning methodology for the flight phase. The core idea here is to change the momentum of the flywheel to control the configuration of the lower body of the robot. This is due to the conservation of the angular centroidal momentum of the robot in the flight phase. Additionally, we add two components to assist the flipping motion with a view toward landing the jump.

### A. Momentum Transmission on the Flywheel

As the flight phase is underactuated due to the conservation of momentum, direct controlling the rotation of the floating robot is not possible. However, it is possible to change the momentum of the lower body of the robot (linkages except the flywheel) by controlling the momentum of the flywheel. The total centroidal pitch momentum  $\mathcal{H}_{\text{pitch}}$  of the robot in the flight phase is,

$$\mathcal{H}_{\text{pitch}} = I_{\text{flywheel}} \omega_{\text{flywheel}} + \sum_{i=2}^N I_i \omega_i, \quad (11)$$

where  $I_{\text{flywheel}}, \omega_{\text{flywheel}}$  are the pitch inertia and velocity of the flywheel about the COM of the robot, and  $I_i, \omega_i$  are the pitch inertia and velocity of the other linkages about the COM. The conservation of momentum indicates  $\mathcal{H}_{\text{pitch}}$  is constant. The averaged angular velocity of lower body can be defined as:

$$\bar{\omega} = \frac{\sum_{i=2}^N I_i \omega_i}{\sum_{i=2}^N I_i}, \quad (12)$$

我们假设机器人以与离地时相似的自配置状态着陆。因此，飞行阶段的持续时间由下式给出：

$$T_{\text{Flight}} = \frac{2\dot{z}^{\text{lift-off}}}{g}, \quad (7)$$

其中  $\dot{z}^{\text{lift-off}}$  是质点离地时的垂直速度。如果机器人在飞行阶段保持构型固定，则离地时全身的角速度应为：

$$\dot{\theta}^{\text{lift-off}} = \frac{\pm 2\pi - 2\beta^{\text{lift-off}}}{T_{\text{Flight}}} = \frac{(\pm\pi - \beta^{\text{lift-off}})g}{\dot{z}^{\text{lift-off}}}. \quad (8)$$

注意，我们稍后将放宽机器人构型固定的条件。式 (8) 建立了跳跃阶段产生翻转运动的约束。符号  $\pm$  取决于空翻是后空翻还是前空翻。

**物理约束：**在跳跃过程中，我们通过以下方式对足部施加防滑条件：

$$-F_z\mu \leq F_x \leq F_z\mu, \quad (\text{摩擦锥})$$

其中  $F_z$  是垂直地面反作用力（GRF）， $\mu$  是摩擦系数。同时，保持足部平贴地面也是可取的，因此引入零力矩点（ZMP）约束：

$$-F_z \frac{l}{2} \leq u \leq F_z \frac{l}{2}, \quad (\text{ZMP})$$

其中  $l$  是足长。其他约束包括驱动限制、腿长限制以及非负法向地面反作用力。最终的地面法向力应降至 0 以触发离地事件，即  $F_z^{\text{lift-off}} = 0$ 。

**成本：**跳跃优化的成本在于虚拟消耗的能量：

$$J_{\text{Jumping}} = \sum_i (c_1 \ddot{L}_i^2 + c_2 u_i^2 + c_3 \tau_i^2) \Delta T, \quad (9)$$

其中  $c_{1,2,3}$  为系数， $\Delta T$  为时间离散化参数， $i$  为离散化变量的索引。图4展示了跳跃行为优化轨迹的一个示例。

### C. 着陆优化

着陆优化在飞行阶段结束后的触地事件之后执行。由于它使用相同的飞轮SLIP模型，我们在此简要说明着陆规划。

在着陆过程中，系统从飞行阶段过渡到静态构型。该任务可直接通过最终状态的等式约束进行编码：

$$\begin{aligned} z_N &= z^{\text{des}}, \\ \dot{z}_N &= x_N = \dot{x}_N = \dot{\theta} = 0, \\ F_N^s &= mg, \end{aligned}$$

其中下标  $N$  表示着陆的最终状态。物理约束与跳跃规划相同。

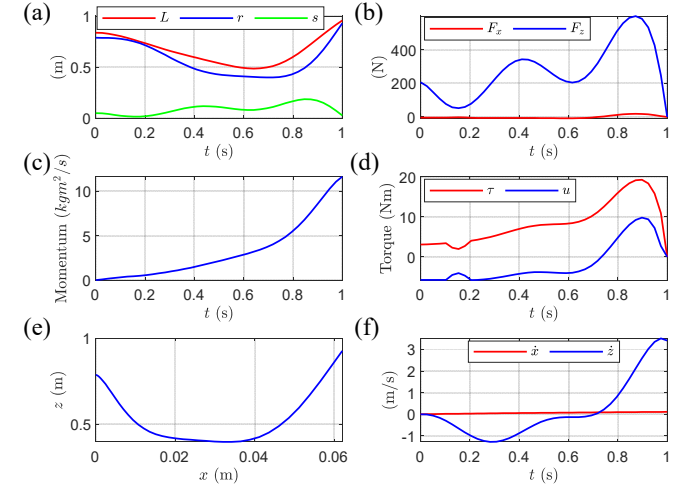


图4. 优化跳跃轨迹示例：腿长与弹簧变形 (a)、地面反作用力 (b)、飞轮动量 (c)、力矩 (d)、质点位置 (e) 及质点速度 (f)。

初始状态根据机器人碰撞后的状态进行约束。代价函数为跳跃代价加上弹簧振荡的惩罚项：

$$J_{\text{Landing}} = J_{\text{Jumping}} + \sum_i \alpha \dot{s}_i^2 \Delta T, \quad (10)$$

其中  $\alpha$  是一个加权系数。

## IV. 飞行阶段规划

在本节中，我们介绍飞行阶段的规划方法。其核心思想是通过改变飞轮的动量来控制机器人下半身的构型。这是因为在飞行阶段，机器人的质心角动量守恒。此外，我们增加了两个辅助翻转运动的组件，以期实现跳跃着陆。

### A. 飞轮动量传递

由于飞行阶段因动量守恒而处于欠驱动状态，因此无法直接控制浮动机器人的旋转。然而，可以通过控制飞轮的动量来改变机器人下半身（除飞轮外的连杆）的动量。飞行阶段机器人的总质心俯仰动量  $\mathcal{H}_{\text{pitch}}$  为：

$$\mathcal{H}_{\text{pitch}} = I_{\text{flywheel}} \omega_{\text{flywheel}} + \sum_{i=2}^N I_i \omega_i, \quad (11)$$

其中  $I_{\text{flywheel}}, \omega_{\text{flywheel}}$  是飞轮相对于机器人质心的俯仰惯量和角速度，而  $I_i, \omega_i$  是其他连杆相对于质心的俯仰惯量和角速度。动量守恒表明  $\mathcal{H}_{\text{pitch}}$  是恒定的。下半身的平均角速度可定义为：

$$\bar{\omega} = \frac{\sum_{i=2}^N I_i \omega_i}{\sum_{i=2}^N I_i}, \quad (12)$$

which is controllable by changing the momentum of the flywheel. If we assume the robot is rigid, the flipping condition becomes,

$$\int_0^{T_{\text{Flight}}} \bar{\omega}(t) = \pm 2\pi - 2\beta^{\text{lift-off}}. \quad (13)$$

The initial pitch angle of the lower body is  $\theta(t=0) = \beta^{\text{lift-off}}$ , and the final angle at touch-down is  $\theta(t=T_{\text{Flight}}) = \pm 2\pi - \beta^{\text{lift-off}}$ , where  $\pm$  depends on the direction of the somersault. The initial velocity  $\bar{\omega}^{\text{lift-off}}$  is calculated from the robot, and we let the final velocity  $\bar{\omega}^{\text{touch-down}} = 0$ . Thus, trajectories of  $[\theta^{\text{des}}(t), \bar{\omega}^{\text{des}}(t)]$  can be constructed based on the boundary conditions. Since it is desirable to adjust the pitch angle of the lower body to prepare for landing at good configurations, we add a feedback law on the desired average velocity:

$$\dot{\theta}_{cl}^{\text{des}}(t) = \bar{\omega}^{\text{des}}(t) - k_p(\theta_{\text{pelvis}}(t) - \theta^{\text{des}}(t)), \quad (14)$$

where we will use the pelvis pitch angle  $\theta_{\text{pelvis}}$  as a reference for the orientation of the lower body. Thus we control the pelvis pitch angle to be the same of the leg angle  $\beta$  on the ground phase. Finally, the desired centroidal momentum of the flywheel is:

$$\mathcal{H}_{\text{flywheel}}^{\text{des}}(t) = \mathcal{H}_{\text{pitch}} - \dot{\theta}_{cl}^{\text{des}}(t)(\sum_{i=2}^N I_i). \quad (15)$$

The rotational momentum required to achieve the whole body rotation is mainly accumulated on the flywheel during the jumping phase. In the flight phase, the momentum is transmitted from the flywheel to the lower body to achieve the desired rotation. Thus we refer Eq. (15) as the *momentum transmission* of the flywheel.

#### B. Assistive Components

In addition to the control of the flywheel, the lower body is also controlled to assist the somersault. We consider two assistive components as follows:

**Leg Retraction:** The legs of the robot can retract in the beginning of the flight phase and then extend towards the landing. This is referred to as *tucking* in [2]. The purpose of the tucking is to decrease the centroidal inertia and thus increase the angular rotational rate. We implement this strategy by controlling the leg length. The desired trajectory is constructed smoothly with appropriate timing.

**Foot Placement:** The robot is expected to achieve the desired angular rotation and then land on the ground. To have the appropriate landing configuration, the robot should control the horizontal positions of the feet, which is controlled by adjusting the leg angle to the opposite of the leg angle at lift-off. By symmetry, the opposite leg angle can ensure the feasibility of landing in the optimization. Additionally, we control the feet to be flat to prepare for landing.

### V. OUTPUT CONSTRUCTION AND STABILIZATION

In this section, we explain the control of the aforementioned plans as they are realized on the 3D robot Cassie. We first explain the output selection and construction in each domain, then derive the optimization-based control method for stabilization.

In our previous work [28], [34], we applied the control Lyapunov function based quadratic programs (CLF-QPs) [35] for output stabilization with momentum outputs. A different optimization-based control, i.e., task space control (TSC) [36], [37], [38], [39], can serve the same purpose of stabilization. Here we derive TSC with minimum variable formulation for illustration.

#### A. Output Construction

The output construction is used to specify the desired behavior from the Flywheel SLIP model and the momentum transmission for the robot via featured functions (outputs) of the states. There are different outputs to be specified in the ground and flight phases.

1) *Jumping/Landing:* The Flywheel SLIP provides the desired point mass dynamics and rotational dynamics for the robot. The point mass dynamics in the leg axial direction is enforced by the leg length  $L_{\text{robot}}(q)$  on the robot; the mass dynamics in the translational/forward direction is enforced by the COM forward position  $x_{\text{com}}(q)$  of the robot. The rotational dynamics is enforced via the centroidal pitch momentum  $\mathcal{H}_{\text{pitch}}(q, \dot{q})$  of the robot. Additionally, we enforce zero motion in the coronal and transversal planes by constraining the lateral position of the COM  $y_{\text{com}}(q) \rightarrow 0$  and the yaw angle of the pelvis  $\phi_{\text{yaw}}(q) \rightarrow 0$ . The outputs are the same for the jumping and landing phase, which are:

$$\mathcal{Y}_1^{J/L}(q, \dot{q}, t) = \mathcal{H}_{\text{Pitch}}(q, \dot{q}) - \mathcal{H}_{\text{FSLIP}}(t), \quad (16)$$

$$\mathcal{Y}_2^{J/L}(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ x_{\text{com}}(q) \\ y_{\text{com}}(q) \\ \phi_{\text{pitch}}(q) \\ \phi_{\text{yaw}}(q) \end{bmatrix} - \begin{bmatrix} L_{\text{FSLIP}}(t) \\ x_{\text{FSLIP}}(t) \\ 0 \\ \phi_{\text{pitch}}^{\text{des}}(t) \\ 0 \end{bmatrix}. \quad (17)$$

where  $\mathcal{H}_{\text{FSLIP}}(t) = I_{\text{FSLIP}}(t)\theta(t)$  is the momentum of the Flywheel SLIP,  $x_{\text{FSLIP}}(t)$  is the forward position of the Flywheel SLIP, and  $\phi_{\text{pitch}}^{\text{des}}(t)$  is the desired pitch angle of the pelvis by construction. The subscript  $1/2$  indicates the number of times one must differentiate to have the torque  $u$  appear.

2) *Flight:* As mentioned in the previous section, the flight phase is underactuated. The pitch momentum is transmitted between the flywheel and the lower body by Eq. (15). Thus the momentum of the flywheel  $\mathcal{H}_{\text{flywheel}}(q, \dot{q})$  is controlled:

$$\mathcal{Y}_1^F(q, \dot{q}) = \mathcal{H}_{\text{flywheel}}(q, \dot{q}) - \mathcal{H}_{\text{flywheel}}^{\text{des}}(t). \quad (18)$$

To enable body tucking, the leg length  $L_{\text{robot}}(q)$  are controlled to retract first and then extend. The rest joints of the body  $q_{/\text{knee}}$ , i.e. motor joints except the knee joints, are controlled as fixed. Thus the outputs are defined as:

$$\mathcal{Y}_2^F(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ q_{/\text{knee}} \end{bmatrix} - \begin{bmatrix} L^{\text{des}}(t) \\ q_{/\text{knee}}^+ \end{bmatrix}, \text{ if } t < T_1 \quad (19)$$

where  $T_1$  is manually decided, e.g.,  $T_1 = \frac{3}{4}T_{\text{Flight}}$ . As approaching landing, it is desirable to prepare for the touch-down event by controlling the leg angle  $\beta_{\text{leg}}(q)$  and toe pitch

这可以通过改变飞轮的动量来控制。如果假设机器人是刚性的，则翻转条件变为：

$$\int_0^{T_{\text{Flight}}} \bar{\omega}(t) = \pm 2\pi - 2\beta^{\text{lift-off}}. \quad (13)$$

下半身的初始俯仰角为  $\theta(t=0) = \beta^{\text{lift-off}}$ ，触地时的最终角度为  $\theta(t=T_{\text{Flight}}) = \pm 2\pi - \beta^{\text{lift-off}}$ ，其中  $\pm$  取决于空翻的方向。初始速度  $\bar{\omega}^{\text{lift-off}}$  根据机器人计算得出，我们设最终速度为  $\bar{\omega}^{\text{touch-down}} = 0$ 。因此，可以基于边界条件构建  $[\theta^{\text{des}}(t), \bar{\omega}^{\text{des}}(t)]$  的轨迹。由于需要调整下半身的俯仰角以在良好构型下准备着陆，我们在期望平均速度上添加了一个反馈律：

$$\dot{\theta}_{cl}^{\text{des}}(t) = \bar{\omega}^{\text{des}}(t) - k_p(\theta_{\text{pelvis}}(t) - \theta^{\text{des}}(t)), \quad (14)$$

其中我们将使用骨盆俯仰角  $\theta_{\text{pelvis}}$  作为下半身朝向的参考。因此，在地面阶段，我们控制骨盆俯仰角与腿角  $\beta$  保持一致。最后，飞轮的期望质心动量为：

$$\mathcal{H}_{\text{flywheel}}^{\text{des}}(t) = \mathcal{H}_{\text{pitch}} - \dot{\theta}_{cl}^{\text{des}}(t)(\sum_{i=2}^N I_i). \quad (15)$$

实现全身旋转所需的旋转动量主要在跳跃阶段积累于飞轮上。在飞行阶段，动量从飞轮传递至下半身以实现期望的旋转。因此，我们将式(15)称为飞轮的动量传递。

#### B. 辅助组件

除了对飞轮的控制外，还对下半身进行控制以辅助空翻。我们考虑以下两种辅助组件：

**腿部收缩：**机器人的腿部可在飞行阶段开始时收缩，然后在着陆前伸展。这在 [2]中被称为收腿。收腿的目的是减小质心惯性，从而提高角旋转速率。我们通过控制腿长来实现这一策略。期望轨迹以适当的时机平滑构建。

**足部放置：**机器人需实现期望的角旋转并随后着陆于地面。为获得合适的着陆构型，机器人应控制足部的水平位置，这通过调整腿角至离地时腿角的对侧角度来实现。由对称性可知，对侧腿部角度可确保优化中着陆的可行性。此外，我们控制足部保持平坦以准备着陆。

### V. 输出构建与稳定化

本节中，我们将阐述上述规划在3D机器人Cassie上实现时的控制方法。首先解释各域中的输出选择与构建，随后推导用于稳定化的基于优化的控制方法。

在先前的工作 [28], [34], 中，我们应用了基于控制李雅普诺夫函数的二次规划 (CLF-QPs)[35] 来实现动量输出的输出稳定化。另一种基于优化的控制方法，即任务空间控制 (TSC) [36], [37], [38], [39],，可达到相同的稳定化目的。此处我们采用最小变量公式推导TSC以作说明。

#### A. 输出构建

输出构建用于通过状态的特征函数（输出）来指定飞轮SLIP模型和动量传递所期望的行为。在地面阶段和飞行阶段需要指定不同的输出。

1) 跳跃/着陆：飞轮SLIP为机器人提供了期望的质点动力学和旋转动力学。机器人腿部轴向的质点动力学由腿长  $L_{\text{robot}}(q)$ 强制执行；平动/前向方向的质量动力学由机器人的质心前向位置  $x_{\text{com}}(q)$ 强制执行。旋转动力学通过机器人的质心俯仰动量  $\mathcal{H}_{\text{pitch}}(q, \dot{q})$ 强制执行。此外，我们通过约束质心横向位置  $y_{\text{com}}(q) \rightarrow 0$ 和骨盆偏航角  $\phi_{\text{yaw}}(q) \rightarrow 0$ 来强制冠状面和横截面上的零运动。跳跃阶段和着陆阶段的输出相同，具体如下：

$$\mathcal{Y}_1^{J/L}(q, \dot{q}, t) = \mathcal{H}_{\text{Pitch}}(q, \dot{q}) - \mathcal{H}_{\text{FSLIP}}(t), \quad (16)$$

$$\mathcal{Y}_2^{J/L}(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ x_{\text{com}}(q) \\ y_{\text{com}}(q) \\ \phi_{\text{pitch}}(q) \\ \phi_{\text{yaw}}(q) \end{bmatrix} - \begin{bmatrix} L_{\text{FSLIP}}(t) \\ x_{\text{FSLIP}}(t) \\ 0 \\ \phi_{\text{pitch}}^{\text{des}}(t) \\ 0 \end{bmatrix}. \quad (17)$$

其中  $\mathcal{H}_{\text{FSLIP}}(t) = I_{\text{FSLIP}}(t)\theta(t)$  是飞轮弹簧负载倒立摆的动量， $x_{\text{FSLIP}}(t)$  是飞轮弹簧负载倒立摆的前向位置，而  $\phi_{\text{pitch}}^{\text{des}}(t)$  是通过构造得到的期望骨盆俯仰角。下标  $1/2$  表示需要微分的次数，以使扭矩  $u$  出现。

2) 飞行阶段：如前一节所述，飞行阶段是欠驱动的。俯仰动量通过公式 (15) 在飞轮与下半身之间传递。因此，飞轮动量  $\mathcal{H}_{\text{flywheel}}(q, \dot{q})$  受到控制：

$$\mathcal{Y}_1^F(q, \dot{q}) = \mathcal{H}_{\text{flywheel}}(q, \dot{q}) - \mathcal{H}_{\text{flywheel}}^{\text{des}}(t). \quad (18)$$

为了实现身体蜷缩，腿长  $L_{\text{robot}}(q)$  被控制为先收缩再伸展。身体的其他关节  $q_{/\text{knee}}$ ，即除膝关节外的电机关节，被控制为固定状态。因此，输出定义为：

$$\mathcal{Y}_2^F(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ q_{/\text{knee}} \end{bmatrix} - \begin{bmatrix} L^{\text{des}}(t) \\ q_{/\text{knee}}^+ \end{bmatrix}, \text{ if } t < T_1 \quad (19)$$

其中  $T_1$  是手动决定的，例如  $T_1 = \frac{3}{4}T_{\text{Flight}}$ 。在接近着陆时，通过控制腿角  $\beta_{\text{leg}}(q)$  和脚趾俯仰来为触地事件做准备是可取的

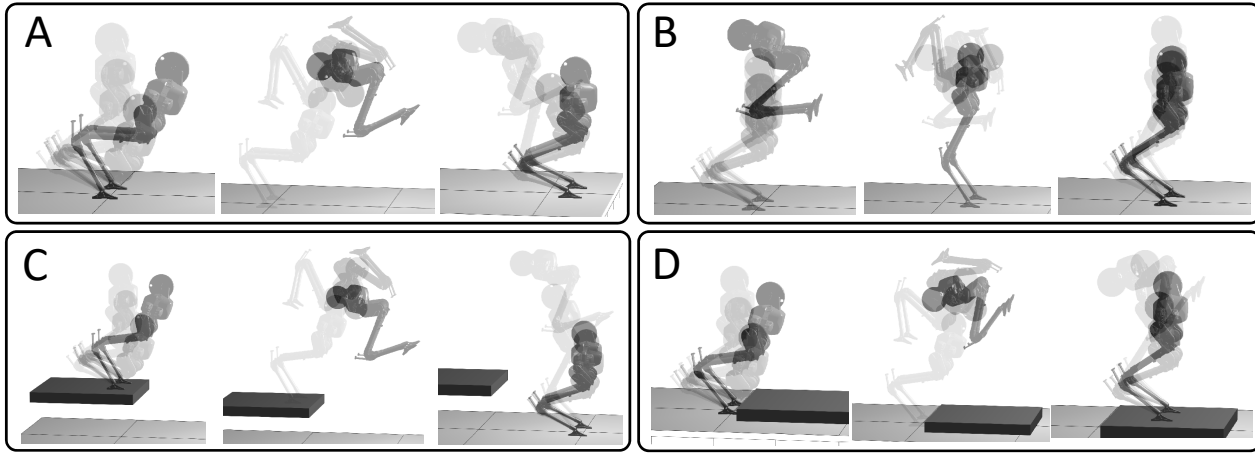


Fig. 5. Snapshots of the generated somersaults: (A) a frontflip with a 1m forward target distance, (B) a backflip, (C) a frontflip from a platform of 0.5m height, and (D) a frontflip without knowing the landing surface, which is with a 0.1m height from the ground.

angle  $q_{\text{toe}}^{\text{pitch}}(q)$ . The outputs are:

$$\mathcal{Y}_2^F(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ \beta_{\text{leg}}^+(q) \\ q_{\text{toe}}^{\text{pitch}}(q) \\ q_{\text{rest}} \end{bmatrix} - \begin{bmatrix} L^{\text{des}}(t) \\ -\beta_{\text{leg}}^+ \\ 0 \\ q_{\text{rest}}^+ \end{bmatrix}, \text{ if } t \geq T_1, \quad (20)$$

where  $\beta_{\text{leg}}^+$  is the leg angle at the lift-off event.  $q_{\text{rest}}$  denotes the rest motor joints except for the constrained one, which in this case are the hip roll and yaw joints.

### B. Momentum-Included Task Space Control

Finally we present the output stabilization via the Task Space Control (TSC) to drive  $\mathcal{Y}_2 \rightarrow 0, \mathcal{Y}_1 \rightarrow 0$ . Typically,

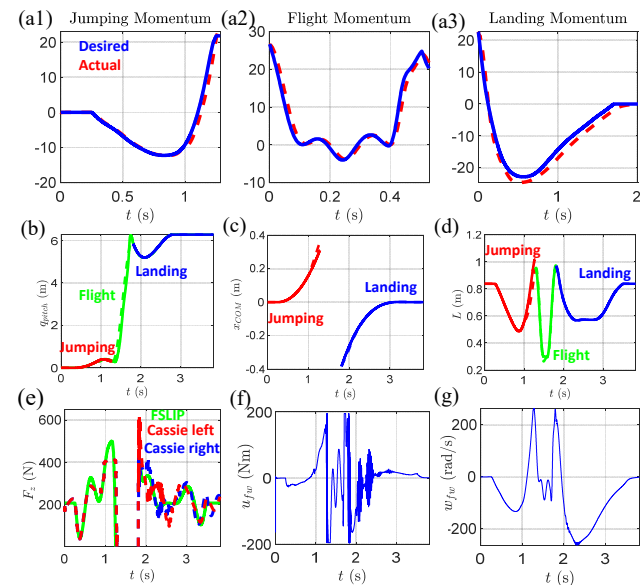


Fig. 6. Simulation plots of the forward frontflip A (dashed lines are the actual; solid lines are the desired): (a1-a3) The momentum output trajectories in each domain (The flywheel momentum is the output in (a2)). (b) The leg length trajectories. (c) The ground norm forces of Cassie v.s. that of the FSLIP. (d)  $x_{\text{COM}}$  (e) The pitch angle of the pelvis. (f) The applied torque on the flywheel. (g) The rotational speed of the flywheel.

the joint accelerations, motor torques, and ground reaction forces are all included as optimization variables in the TSC formulation. Here we present the TSC formulation with only the motor torques as the optimization variables, which lets the TSC based quadratic program be solved efficiently.

First, we notice that the ground reaction forces are affine functions of the input torques  $u$  as:

$$F_v = A_v u + b_v, \quad (21)$$

$$A_v = -(J_v M^{-1} J_v^T)^{-1} J_v M^{-1} B, \quad (22)$$

$$b_v = (J_v M^{-1} J_v^T)^{-1} (J_v M^{-1} H - \dot{J}_v \dot{q}). \quad (23)$$

Plugging this into the equation of motion yields,

$$M(q)\ddot{q} + \underbrace{H(q, \dot{q}) - J_s^T \tau_s(q) - J_{h,v}^T b_v}_{\bar{H}} = \underbrace{(B + J_{h,v}^T A_v)}_{\bar{B}_v} u,$$

which yields the affine relation  $\ddot{q} = \bar{A}u + \bar{b}$  with

$$\bar{A} = M^{-1} \bar{B}_v, \bar{b} = -M^{-1} \bar{H}. \quad (24)$$

As the derivatives of outputs are affine w.r.t.  $\ddot{q}$ , they are also affine w.r.t. to  $u$ :

$$\dot{\mathcal{Y}}_1 = A_1 \ddot{q} + b_1 = \mathcal{A}_1 u + \mathcal{B}_1, \quad (25)$$

$$\dot{\mathcal{Y}}_2 = A_2 \ddot{q} + b_2 = \mathcal{A}_2 u + \mathcal{B}_2, \quad (26)$$

where the explicit expression of each term is omitted. Compactly,

$$\dot{\mathbf{Y}} = \mathcal{A}u + \mathcal{B} \quad (27)$$

with  $\mathbf{Y} = [\mathcal{Y}_1; \mathcal{Y}_2]$ . The desired derivatives (accelerations) for stabilization are,

$$\dot{\mathcal{Y}}_1^{\text{des}} = -K_1^p \mathcal{Y}_1, \quad (28)$$

$$\dot{\mathcal{Y}}_2^{\text{des}} = -K_2^p \mathcal{Y}_2 - K_2^d \dot{\mathcal{Y}}_2. \quad (29)$$

This motivates a quadratic cost to minimize the difference between the actual and desired accelerations:

$$J_{\text{TSC}}(u) = (\mathcal{A}u + \mathcal{B} - \dot{\mathcal{Y}}^{\text{des}})^T (\mathcal{A}u + \mathcal{B} - \dot{\mathcal{Y}}^{\text{des}}), \quad (30)$$

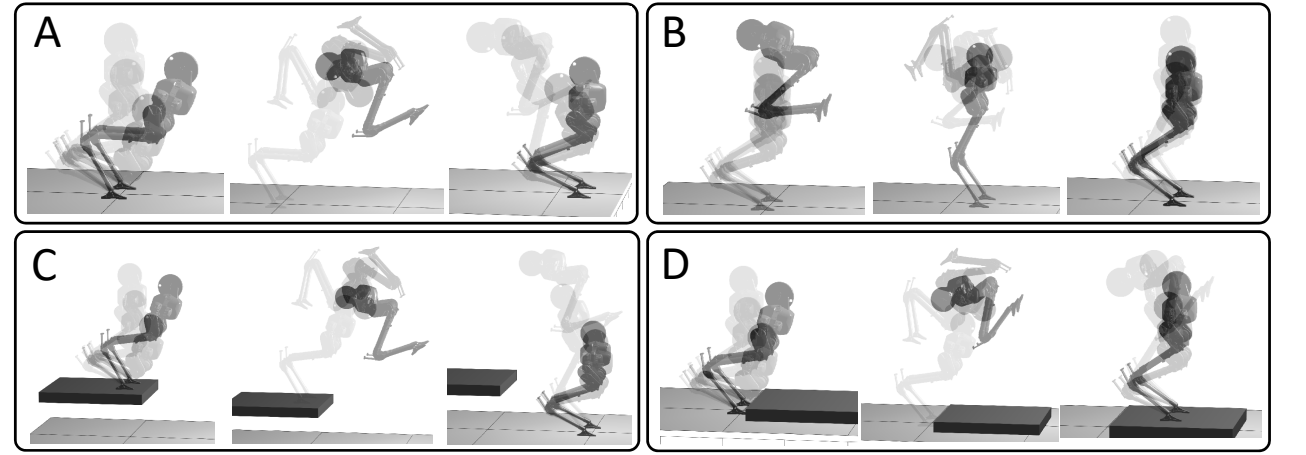


图5. 生成空翻的截图：(A) 前空翻，目标向前距离1米，(B) 后空翻，(C) 从0.5米高平台进行的前空翻，以及 (D) 在未知情着陆面情况下进行的前空翻，该着陆面距地面高度为0.1米。

角度  $q_{\text{toe}}^{\text{pitch}}$  脚趾  $(q)$ 。输出为：

$$\mathcal{Y}_2^F(q, t) = \begin{bmatrix} L_{\text{robot}}(q) \\ \beta_{\text{leg}}^+(q) \\ q_{\text{toe}}^{\text{pitch}}(q) \\ q_{\text{rest}} \end{bmatrix} - \begin{bmatrix} L^{\text{des}}(t) \\ -\beta_{\text{leg}}^+ \\ 0 \\ q_{\text{rest}}^+ \end{bmatrix}, \text{ if } t \geq T_1, \quad (20)$$

其中  $\beta^+$  腿角是离地事件时的腿角。  $q_{\text{rest}}$  表示除受约束关节外的其余电机关节，在本例中为髋关节的横滚和偏航关节。

### B. 含动量的任务空间控制

最后，我们通过任务空间控制 (TSC) 实现输出稳定化，以驱动  $\mathcal{Y}_2 \rightarrow 0, \mathcal{Y}_1 \rightarrow 0$ 。通常，

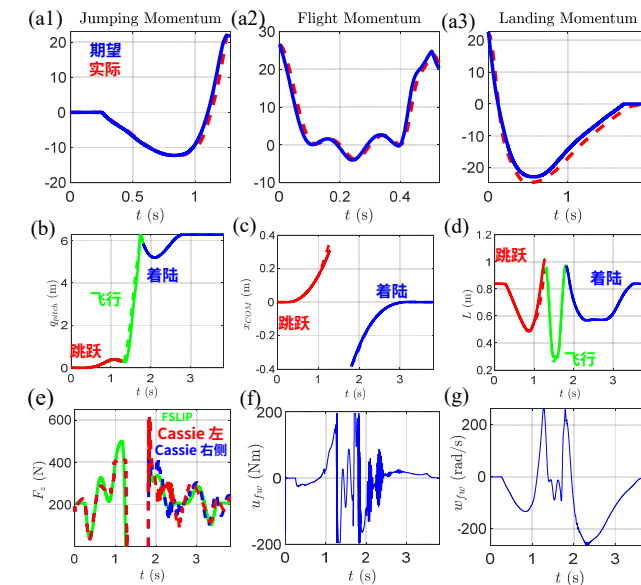


图6. 前向空翻A的仿真图（虚线为实际值；实线为期望值）：(a1-a3) 各域中的动量输出轨迹（(a2)中输出为飞轮动量）。(b) 腿长轨迹。(c) Cassie的地面法向力与FSLIP的地面法向力对比。(d)  $x_{\text{COM}}$  (e) 骨盆俯仰角。(f) 施加在飞轮上的扭矩。(g) 飞轮的转速。

关节加速度、电机扭矩和地面反作用力都被纳入TSC公式中的优化变量。在此，我们提出仅以电机扭矩作为优化变量的TSC公式，这使得基于TSC的二次规划能够高效求解。

首先，我们注意到地面反作用力是输入力矩  $u$  的仿射函数：

$$F_v = A_v u + b_v, \quad (21)$$

$$A_v = -(J_v M^{-1} J_v^T)^{-1} J_v M^{-1} B, \quad (22)$$

$$b_v = (J_v M^{-1} J_v^T)^{-1} (J_v M^{-1} H - \dot{J}_v \dot{q}). \quad (23)$$

将其代入运动方程可得，

$$M(q)\ddot{q} + \underbrace{H(q, \dot{q}) - J_s^T \tau_s(q) - J_{h,v}^T b_v}_{\bar{H}} = \underbrace{(B + J_{h,v}^T A_v)}_{\bar{B}_v} u,$$

由此得到仿射关系  $\ddot{q} = \bar{A}u + \bar{b}$  其中

$$\bar{A} = M^{-1} \bar{B}_v, \bar{b} = -M^{-1} \bar{H}. \quad (24)$$

由于输出的导数关于  $\ddot{q}$  是仿射的，因此它们关于  $u$  也是仿射的：

$$\dot{\mathcal{Y}}_1 = A_1 \ddot{q} + b_1 = \mathcal{A}_1 u + \mathcal{B}_1, \quad (25)$$

$$\dot{\mathcal{Y}}_2 = A_2 \ddot{q} + b_2 = \mathcal{A}_2 u + \mathcal{B}_2, \quad (26)$$

其中省略了各项的显式表达式。简而言之，

$$\dot{\mathbf{Y}} = \mathcal{A}u + \mathcal{B} \quad (27)$$

其中  $\mathbf{Y} = [\mathcal{Y}_1; \mathcal{Y}_2]$ 。用于稳定化的期望导数（加速度）为，

$$\dot{\mathcal{Y}}_1^{\text{des}} = -K_1^p \mathcal{Y}_1, \quad (28)$$

$$\dot{\mathcal{Y}}_2^{\text{des}} = -K_2^p \mathcal{Y}_2 - K_2^d \dot{\mathcal{Y}}_2. \quad (29)$$

这促使采用二次代价来最小化实际加速度与期望加速度之间的差异：

$$J_{\text{TSC}}(u) = (\mathcal{A}u + \mathcal{B} - \dot{\mathcal{Y}}^{\text{des}})^T (\mathcal{A}u + \mathcal{B} - \dot{\mathcal{Y}}^{\text{des}}), \quad (30)$$



with  $\dot{\mathcal{Y}}^{des} = [\dot{\mathcal{Y}}_1^{\text{des}^T}, \ddot{\mathcal{Y}}_2^{\text{des}^T}]^T$ .

Additionally, two physical constraints are included for the feasibility of the torques on the robots. In the ground phase, the ground reaction forces have to satisfy the physics constraints such as non-negative normal forces and non-slipping condition, formulated by  $C_v F_v \leq 0$ , where  $C_v$  is a constant matrix. The constraint is also on  $u$  from Eq. (21):

$$C_v A_v u \leq -C_v b_v. \quad (31)$$

Moreover, the motor torque must be within the feasible limits of the robot hardware. For electrical motors, the available motor torque depends on the rotational speed. Then we encode that motor torque limit constraint as:

$$u_{lb}(\dot{q}) \leq u \leq u_{ub}(\dot{q}). \quad (32)$$

Thus the quadratic program based controller for each domain is formulated as:

$$\begin{aligned} u^* = \underset{u \in \mathbb{R}^{11}}{\operatorname{argmin}} \quad & J_{\text{TSC}}(u), \\ \text{s.t.} \quad & C_v A_v u \leq -C_v b_v, \quad (\text{GRF}) \\ & u_{lb}(\dot{q}) \leq u \leq u_{ub}(\dot{q}). \quad (\text{Torque Limit}) \end{aligned}$$

## VI. RESULTS AND DISCUSSION

The proposed planning and optimization-based controller are implemented in simulation on the modified robot model of Cassie. The dynamics are integrated using Matlab ODE45 with event triggering. The trajectory optimization is solved via IPOPT [40]. Due to the simple dynamics, the optimizations are typically solved within seconds without code optimization for speed. The TSC is solved at 2kHz using qpOASES [41]. The results can be seen in [42].

Using the proposed method, we generated several somersault motions on the robot. We specify the desired jumping height and forward distance, and then the optimization on Flywheel SLIP is performed. Jumping is thus generated on the robot. When the robot lifts off, the momentum transmission control is applied. After touch-down, the landing optimization is performed. Then the robot lands to a static configuration by tracking the optimized landing trajectory.

Even though the model difference between the Flywheel SLIP and the robot can cause about 0.1m difference on the jumping height, the sequential planning and control still can achieve successful landing and accurate forward distance. Due to the numerical precision of integration, the robot can behave asymmetrically across the frontal place; since lateral COM position is controlled at the ground phase, the robot is still stabilized. Fig. 5 demonstrates four somersaults. Fig. 6 show the trajectories of the produced somersault A.

**Robust to ground height change:** Suppose the ground height of the landing is not exactly known, the robot can still land successfully. The reason is that landing planning is performed after the post-impact event. The somersault D in Fig. 5 is an example where the ground height increases by 0.1m, which shows the robustness of the planning and control with respect to the ground height change.

**Practical Feasibility:** The result in Fig. 6 is on the robot with flywheel I selected in Table I, which has relatively small inertia comparing to the inertia of the rest of the robot. Therefore, the required rotational speed is high, and we replace Eq. (32) on the flywheel by a constant torque constraint to explore the required actuation. As a comparison, the knee motor module of Cassie has a maximum torque of 195.2(Nm); while the maximum speed after gear reduction is only 81.25(rpm). This implies that we need a significant improvement on the actuation design on the flywheel to realize such somersaulting motion.

We also tried different flywheel specifications in Table I to realize somersault C in Fig. 5, with an apex height of 1.2m from the platform. By limiting the maximum joint torque, the maximum required flywheel speed decreases with the increases of the flywheel inertia. However, if the mass increases, the lower body requires an increase of the maximum torques to generate the desired apex height. If the desired apex increases, which leads to a lower required flywheel momentum due to a longer flight duration, the knee actuation has to be increased. Another way to decrease the required momentum generation on the flywheel is to increase the hip pitch rotational speed at lift-off. We found that the contribution to decreasing the flywheel momentum is small unless the maximum hip joint speed is significantly increased. With the above discussion, we conclude that the practical realization of somersaults relies on powerful actuation design and good inertia distribution. This is a challenging topic to study in the field of legged locomotion.

## VII. CONCLUSION AND FUTURE WORK

To conclude, we proposed a sequential planning and control method for generating somersaults on bipedal robots. The planning on the ground phase is based on the Flywheel SLIP model, and that in the flight phase follows the principle of conservation of momentum using momentum transmission. The planned trajectories are extracted and tracked by momentum-included task space control on the robot Cassie.

Future work will be devoted to understanding the necessary components in achieving other types of aerial motions, including periodic hopping, running, and aerial flipping in the transversal and coronal planes. Additionally, it is interesting to study how to generate somersaults on robots without free-rotational joints. From our perspective, the bottleneck for achieving this kind of highly dynamic motion on the physical robot is on the necessity to unify mechanical design with control implementation at the system level. Future work will explore this trade-off with a view toward control-informed robot design for realizing dynamic aerial motion.

TABLE I  
DIFFERENT FLYWHEEL SPECIFICATIONS

| Flywheel | $I_{\text{pitch}}$ (kg · m <sup>2</sup> ) | $m$ (kg) | $\tau_{\text{max}}$ (Nm) | $w_{\text{max}}$ (rpm) |
|----------|-------------------------------------------|----------|--------------------------|------------------------|
| I        | 0.1074                                    | 9.54     | 195                      | 1645                   |
| II       | 0.2148                                    | 19.09    | 195                      | 812                    |
| III      | 0.3222                                    | 28.63    | 195                      | 573                    |

with  $\dot{\mathcal{Y}}^{des} = [\dot{\mathcal{Y}}_1^{\text{des}^T}, \ddot{\mathcal{Y}}_2^{\text{des}^T}]^T$ .

此外，还包含两个物理约束以确保机器人上力矩的可行性。在地面阶段，地面反作用力必须满足物理约束，例如非负法向力和无滑移条件，这些由  $C_v F_v \leq 0$ 表示，其中  $C_v$  是一个常数矩阵。该约束也适用于式(21)中的  $u$ ：

$$C_v A_v u \leq -C_v b_v. \quad (31)$$

此外，电机力矩必须处于机器人硬件的可行范围内。对于电动机而言，可用电机力矩取决于转速。因此，我们将该电机力矩极限约束编码为：

$$u_{lb}(\dot{q}) \leq u \leq u_{ub}(\dot{q}). \quad (32)$$

因此，每个域的基于二次规划的控制器的表述为：

$$\begin{aligned} u^* = \underset{u \in \mathbb{R}^{11}}{\operatorname{argmin}} \quad & J_{\text{TSC}}(u), \\ \text{s.t.} \quad & C_v A_v u \leq -C_v b_v, \quad (\text{GRF}) \\ & u_{lb}(\dot{q}) \leq u \leq u_{ub}(\dot{q}). \quad (\text{Torque Limit}) \end{aligned}$$

## VI. 结果与讨论

所提出的规划与基于优化的控制器在改进的 Cassie机器人模型仿真中实现。动力学采用Matlab ODE45结合事件触发进行积分。轨迹优化通过IPOPT [40]求解。由于动力学模型简单，优化过程通常无需代码优化即可在数秒内完成求解。TSC以2kHz频率通过 qpOASES [41]求解。结果可见于 [42]。

采用所提出的方法，我们在机器人上生成了多个空翻动作。首先指定期望的跳跃高度和前进距离，随后对飞轮弹簧负载倒立摆进行优化，从而在机器人上生成跳跃动作。当机器人离地时，应用动量传输控制。触地后，执行着陆优化。然后，机器人通过跟踪优化的着陆轨迹，着陆至静态构型。

尽管飞轮弹簧负载倒立摆与机器人之间的模型差异可能导致跳跃高度产生约0.1米的偏差，但顺序规划与控制仍能实现成功着陆和准确的前进距离。由于积分的数值精度，机器人在额状面上可能出现不对称行为；但由于侧向质心组合在地面阶段得到控制，机器人仍能保持稳定。图5展示了四次空翻。图6显示了所生成的空翻A的运动轨迹。

**对地面高度变化具有鲁棒性：**假设着陆点的地面高度并非精确已知，机器人仍能成功着陆。原因在于着陆规划是在碰撞后事件之后执行的。图5中的空翻D就是一个例子，其中地面高度增加了0.1米，这展示了规划与控制对地面高度变化的鲁棒性。

**实际可行性：**图6的结果是在采用表I中选定飞轮惯性的机器人上获得的，该飞轮惯性相对于机器人其余部分的惯性较小。因此，所需的转速较高，我们将飞轮上的方程(32)替换为恒定力矩约束，以探索所需的驱动。作为对比，Cassie的膝关节电机模块最大扭矩为195.2（牛·米）；而齿轮减速后的最大转速仅为81.25（转/分）。这意味着我们需要在飞轮的驱动设计上做出重大改进，才能实现此类空翻运动。

我们还尝试了表I中不同的飞轮规格，以实现图5中的后空翻C，其最高点高度为距离平台1.2米。通过限制最大关节扭矩，所需的最大飞轮转速随飞轮惯量的增加而降低。然而，如果质量增加，下半身需要增大最大扭矩以产生期望最高点高度。如果期望最高点高度增加，由于飞行时间延长，所需飞轮动量降低，但膝关节驱动必须增强。另一种减少飞轮所需动量生成的方法是提高离地时的髋关节俯仰转速。我们发现，除非最大髋关节速度显著提高，否则这对降低飞轮动量的贡献很小。通过上述讨论，我们得出结论：空翻的实际实现依赖于强大的驱动设计和良好的惯量分布。这是腿式运动领域一个具有挑战性的研究课题。

## VII. 结论与未来工作

总之，我们提出了一种用于在双足机器人上生成空翻的序列规划与控制方法。地面阶段的规划基于飞轮 SLIP模型，而飞行阶段的规划则遵循动量守恒原理，通过动量传递实现。规划的轨迹被提取出来，并通过含动量的任务空间控制在Cassie机器人上进行跟踪。

未来的工作将致力于理解实现其他类型空中运动所需的必要组件，包括周期性跳跃、奔跑，以及在横向和冠状面上的空中翻转。此外，研究如何在没有自由旋转关节的机器人上生成空翻也颇具意义。从我们的角度来看，在实体机器人上实现此类高度动态运动的瓶颈在于，必须在系统层面将机械设计与控制实现统一起来。未来的工作将探索这一权衡，以期实现控制引导的机器人设计，从而达成动态空中运动。

表I 不同飞轮规格

| 飞轮  | $I_{\text{pitch}}$ (kg · 质量 <sup>2</sup> ) | $m$ (kg) | $\tau_{\text{max}}$ (牛·米) | $w_{\text{max}}$ (转/分) |
|-----|--------------------------------------------|----------|---------------------------|------------------------|
| I   | 0.1074                                     | 9.54     | 195                       | 1645                   |
| II  | 0.2148                                     | 19.09    | 195                       | 812                    |
| III | 0.3222                                     | 28.63    | 195                       | 573                    |

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